

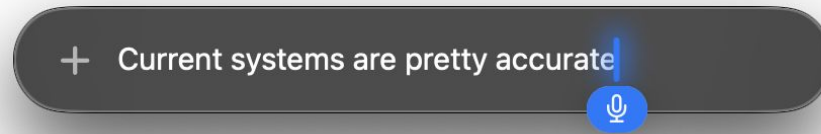
Improving Error Detection for Screenless Text-Dictation Feedback

using Proper Noun Detection



“Sorry, I can’t text now I’m driving”

“Pulchritudinous”



“rheumatoid arthritis”

“I love computer science”

Jesika

Jessica

Tressider

Traitor

+ but sometimes they fall short...

Jon

John

Shawn

Sean, Shaun

Linkin Park

Lincoln Park

How others are improving accuracy (and why transcription will never be perfect)

USER CONTEXT! 🤖

MORE DATA! 💰

BETTER MODELS! 😎

Google

IMPROVING PROPER NOUN RECOGNITION IN END-TO-END ASR BY CUSTOMIZATION OF THE MWER LOSS CRITERION

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Amazon

AWS Machine Learning Blog

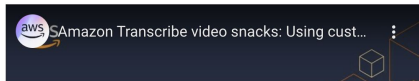
Build a custom vocabulary to enhance speech-to-text transcription accuracy with Amazon Transcribe

by Paul Zhao and Yibin Wang | on 03 JUL 2019 | in [Amazon Transcribe](#), [Artificial Intelligence](#) | [Permalink](#) | [Comments](#) | [Share](#)

[Amazon Transcribe](#) is a fully-managed automatic speech recognition (ASR) service that makes it easy for developers to add speech-to-text capabilities to applications. Depending on your use case, you may have domain-specific terminology that doesn't transcribe properly (e.g. "EBITDA" or "myocardial infarction"). In this post, we will show you how to leverage the custom vocabulary feature – by leveraging custom pronunciations and custom display forms – to enhance transcription accuracy of domain-specific words or phrases that are relevant to your use case.

Custom vocabulary is a powerful feature that helps users transcribe terms that would otherwise not be part of our general ASR service. For instance, your use case may involve brand names or proper names that are not normally part of a language model's regular lexicon, like in the case of "Hogwarts". In this case, it would not only be helpful to be able to add the custom text, but also be able to inform our ASR service on the pronunciation to help our system better recognize unfamiliar terms. On a related note, perhaps you have a term, say, "Lotus" which is a brand name of a car. Naturally, we recognize "lotus" as a flower already. But for your use case, you'd like to have the word transcribed with proper capitalization in the context of recognizing it as a make or model of a vehicle. You can therefore use the recently added custom display forms to achieve this.

First, check out the following video for a demo and then read on to walk through some examples of using both custom pronunciation and also custom display forms.



In recent years, end-to-end (E2E) ASR systems such as RNN-T [10] and LAS [11] have proven to be viable alternatives to conventional speech systems, especially in the on-device setting where memory is limited [12]. These systems achieve strong results with small model size by folding the AM, PM and LM into a single neural model. While this simplifies the model architecture, it removes the explicit PM as a site for the injection of knowledge into the model, making it more difficult to model specific requirements like proper noun pronunciation. It also reduces the exposure of the model to proper nouns, since unlike the LM of a conventional model an E2E system is trained entirely on audio-text pairs.

Several directions have been explored for improving proper noun recognition in E2E models. In [13], phonetic fuzzing of proper nouns in training an E2E system yields improvements on several proper noun benchmarks, but the approach is restricted to the bi-asing case, where candidate proper nouns are passed directly to the model. In addition, several techniques incorporating text-only data into an E2E model, including shallow fusion, cold fusion, and deep fusion have been proposed and have been shown to improve performance on tail utterances [14, 15]. However, this approach requires an external language model to be used in both training and inference.

In an attempt to make up for the lack of an explicit LM potentially trained on billions of words, there has also been work on including large corpora of unsupervised audio or text data during training of an E2E model. The authors of [16] use a conventional model to decode unsupervised utterances that contain proper nouns and train an E2E system on the resulting transcripts. Conversely, the authors of [17] apply a text-to-speech (TTS) system to unsupervised text data and train an E2E system on the resulting audio. In [18] and [19], unsupervised data is included in an E2E model through a separately trained model with a reconstruction loss. While all of these approaches yield improvements, they require use of external datasets.

In this paper, we explore approaches to proper noun recognition

Matthew?

Matthue?

Mathieu?

Mathew?

Math-Yu?



So what?

I can just fix the spelling manually. No biggie, right?

Blind Accessibility

Screenless Interfaces

Existing Technology

Many (biased) **Assumptions;**
Solid Results

***Impossibly* Accurate**
Transcription

Zero Assumptions;
Perfect Results

How do we remove assumptions and bias about the user from the transcription process?

Existing Technology

Many (biased) **Assumptions;**
Solid Results

***Impossibly* Accurate**
Transcription

Zero Assumptions;
Perfect Results

How do we remove assumptions and bias about the user from the transcription process?

User Involvement



Existing Technology

***Impossibly Accurate
Transcription***

**Many (biased) Assumptions;
Solid Results**

**Zero Assumptions;
Perfect Results**





Big Question

Does involving users in the error detection and correction process improve the usability and accuracy of transcription technologies?



Medium Question

What ***IS*** the best way to involve users in the error detection and correction process?



Big Idea

Detecting proper nouns, then presenting them to the user for manual verification + correction.

How we answered it:

User study testing 2 different forms of proper noun feedback against a control group.

Control Group



Record Audio



Transcribe Text



**Play back
Transcription**

Variation 2:

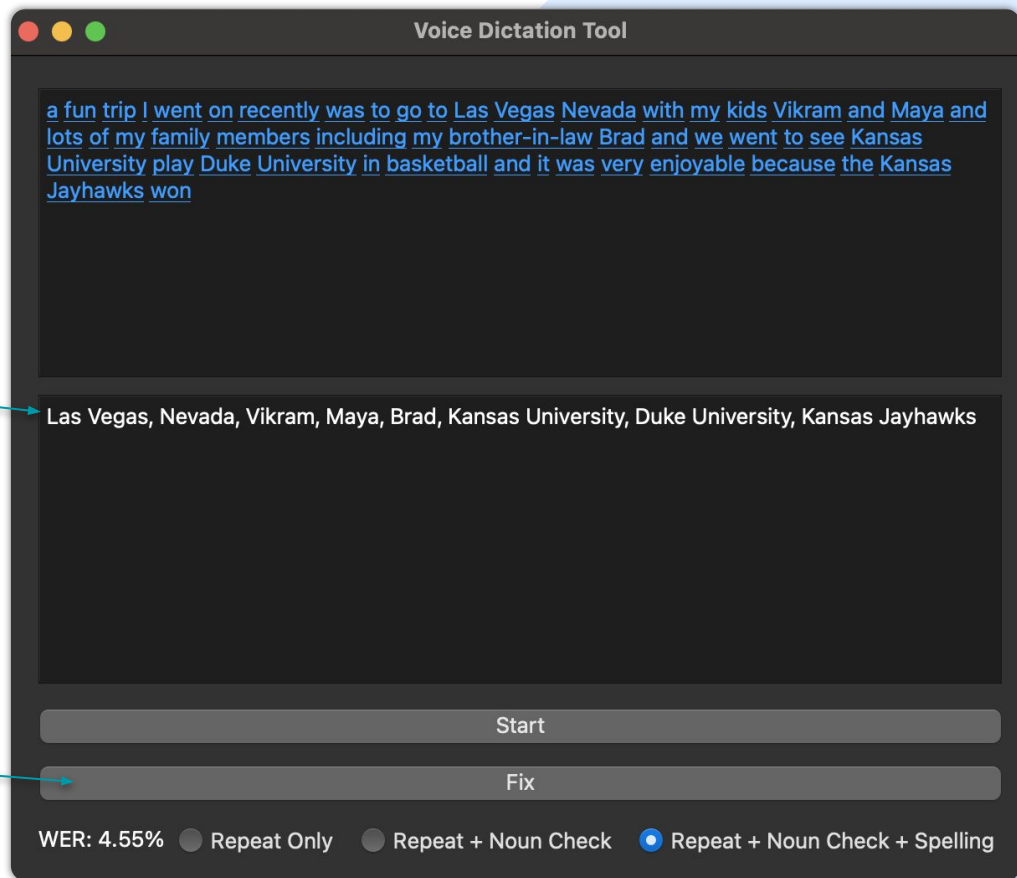
Proper Noun Announcement + Error Correction



Var 2. Interface

Proper-noun
detection

Option to fix
after recording



Variation

Live Transcription + Live Spelling Feedback

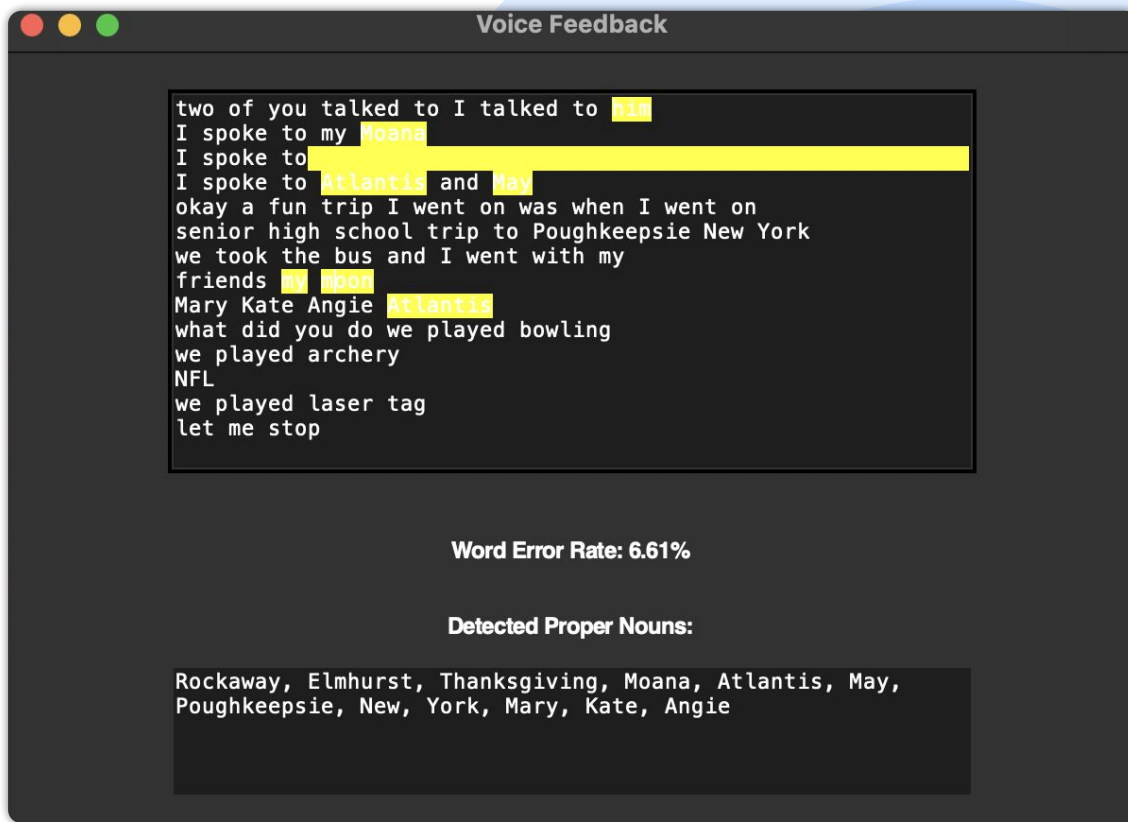
User input

"Hi! My name is Maimuna, and I love CS 197! I can't wait to do more research."



System Feedback

Var 1. Interface

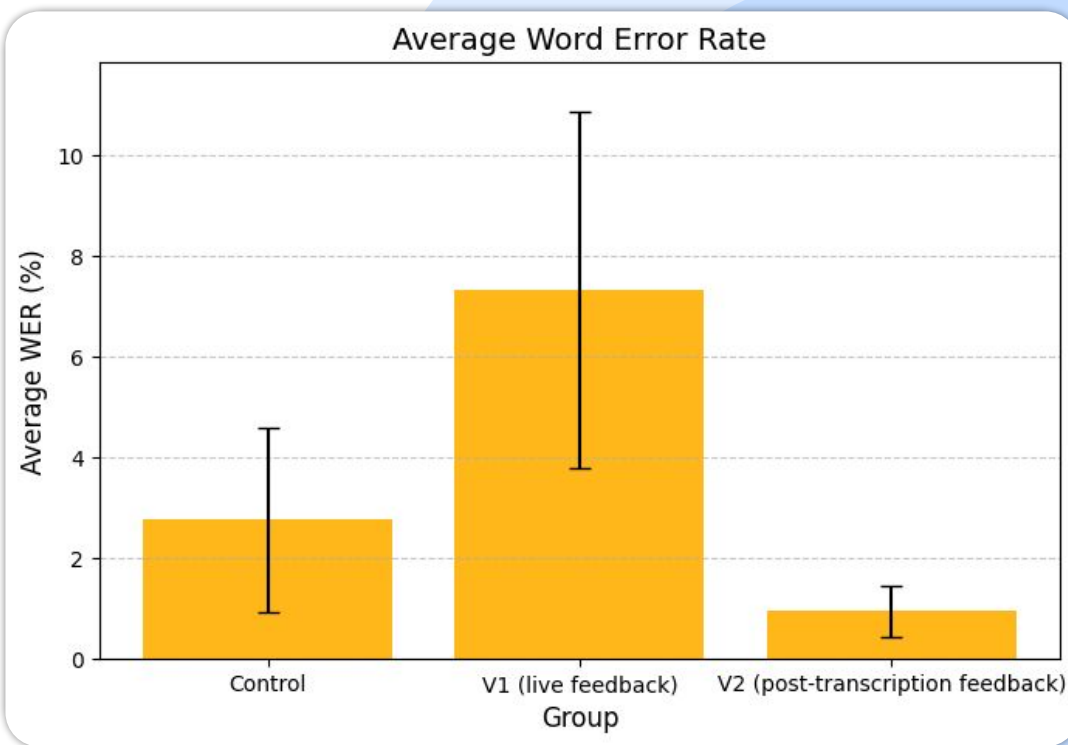


What did we find?

Variation 2

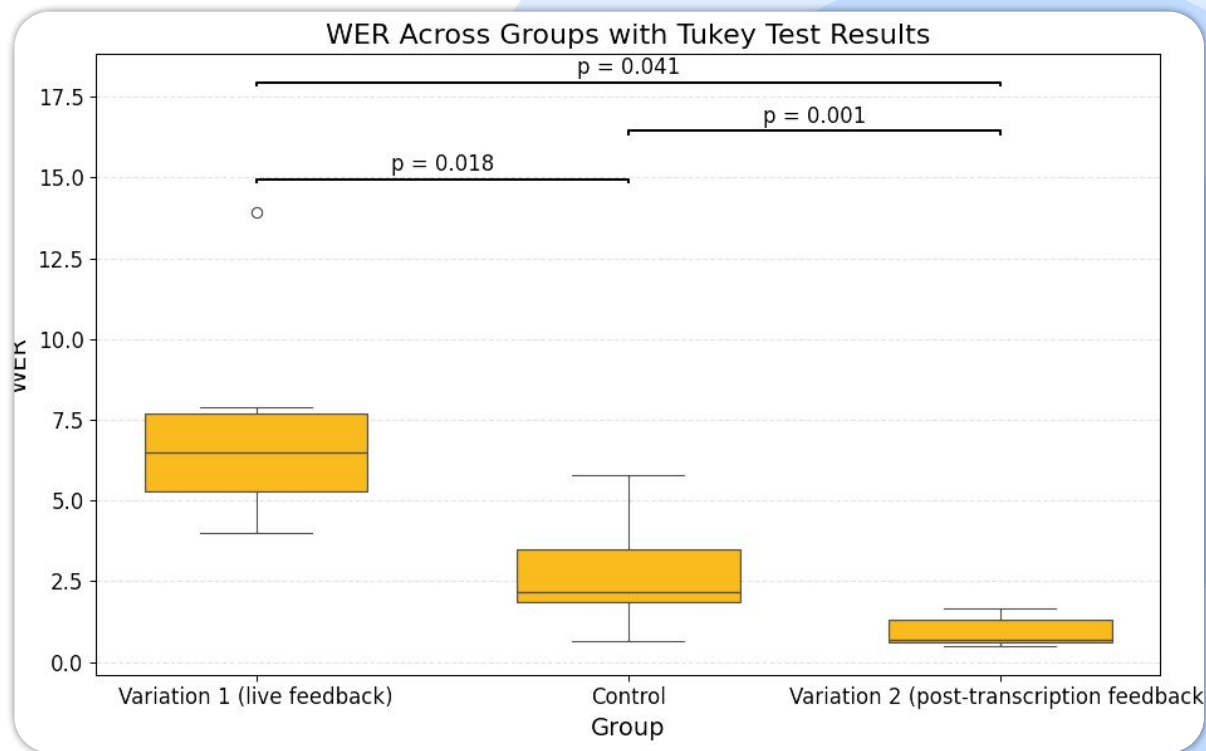
**(post-transcription
feedback)**

had the lowest Word Error
Rate



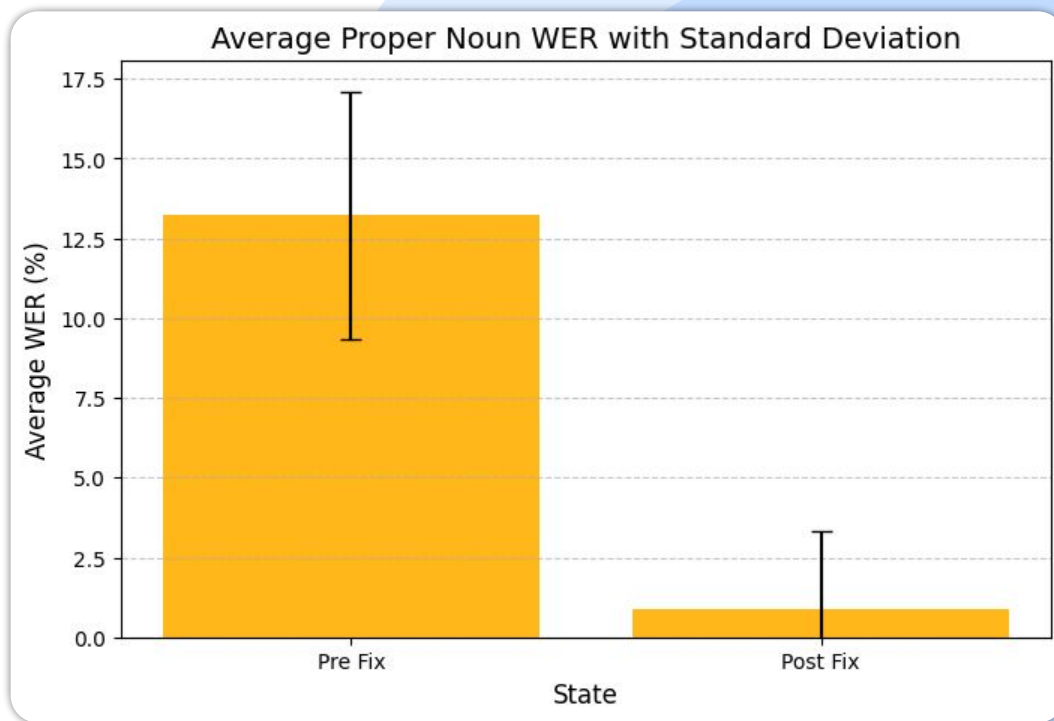
**Statistically
Significant?**

Yep.



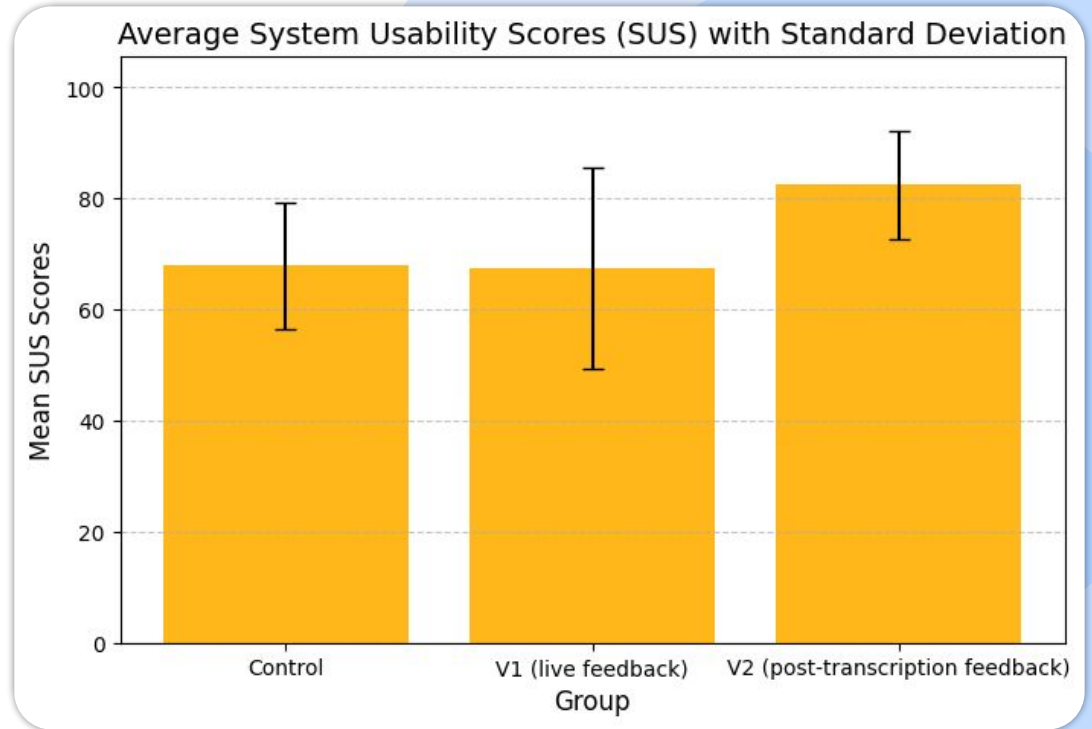
Post-correction,

**Proper Noun WER
decreased... by a lot!**



Usability Scores

Variation 2 out performed the control and V1.



"The control system felt like what I'm used to, but variations 1 and 2 are better for proper nouns."

"The interruptions made it hard to keep my train of thought, which is why I'd prefer Variation 2."

Users preferred non-interrupted speech over live feedback

"Variation 2 allowed me to finish my thoughts and fix mistakes afterward"



Key Takeaways!

Our findings, simplified:

- Users prefer uninterrupted speech
- Integrated options to correct errors increase perceived usability

Implications on other areas:

- UX & HCI Research
- Blind Accessibility
- Screenless Interfaces

Thanks for *listening*

